

Western Mathematics

2022

YEAR 11
FINAL EXAMINATION

Mathematics Standard

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks : **Section I – 10 marks** (pages 2 – 7)
80

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 9 – 32)

- Attempt Questions 11 – 36
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

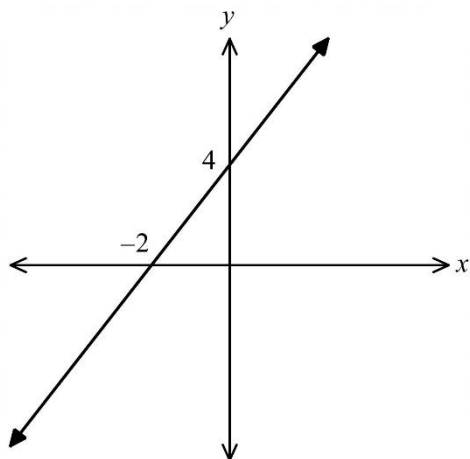
Use the multiple-choice answer sheet for Questions 1 – 10

1. A retailer buys a necklace for \$120.00 and adds 40% to the price for her profit margin.
What is the selling price of the necklace after 10% GST is added to the retailer's price?

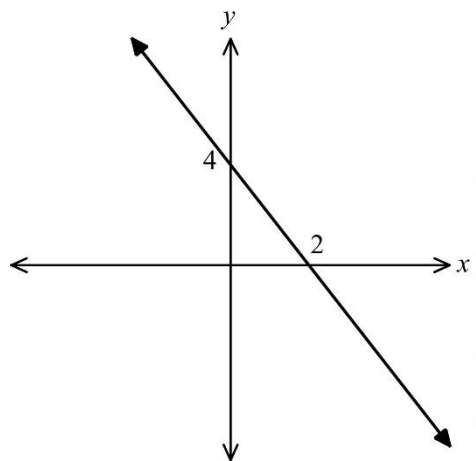
- (A) \$176.00
- (B) \$178.00
- (C) \$180.00
- (D) \$184.80

2. Which of the following graphs is represented by the equation $y = -4x + 2$.

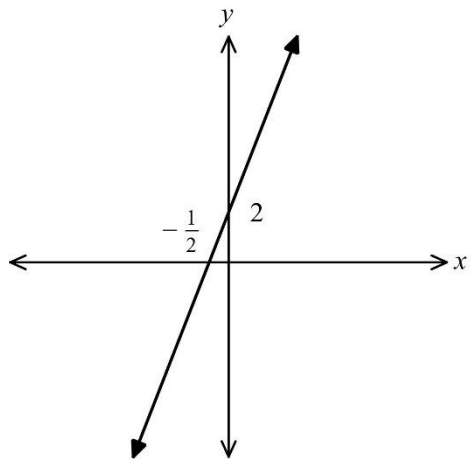
(A)



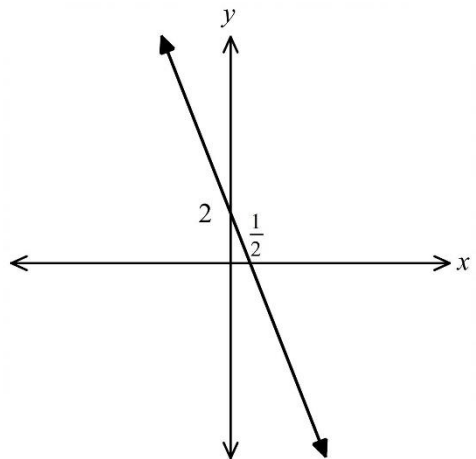
(B)



(C)



(D)



3. Write the number 0.003 567 92 in standard notation, correct to 3 significant figures.

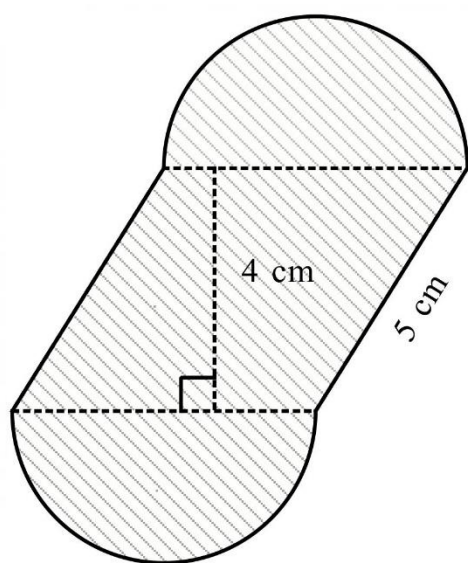
(A) 0.35×10^{-2}

(B) 3.56×10^{-3}

(C) 3.57×10^{-2}

(D) 3.57×10^{-3}

4. The shape shown below consists of a rhombus with a semicircle on two of the opposite sides as shown below.



NOT TO
SCALE

Calculate the area of the shape.

(A) 29.8 cm^2

(B) 39.6 cm^2

(C) 44.6 cm^2

(D) 98.5 cm^2

5. Natalie works in sales and is paid a retainer of \$500 per week, plus a commission. The commission is calculated at 3% of her sales up to and including \$24 000 and at 5% of her sales over \$24 000. What would she earn in a week where her sales totalled \$36 000?
- (A) \$1 460.00
 (B) \$1 580.00
 (C) \$1 820.00
 (D) \$2 300.00
6. The stem-and-leaf plot shows the number of cars that use a restaurant carpark each hour over a 24-hour period.

0	4	5	8	9			
1	1	1	3	5	6	8	
2	2	6	8	8	9	9	9
3	0	2	2	4			
4	0	2	4				

What is the median of this dataset?

- (A) 26
 (B) 27
 (C) 28
 (D) 29

7. An excerpt from Richard's Water Account for April to June is shown below.

Service charges for the period 1st April to 30th June 2022			
	Number of Services	Charge	Amount
Residential Sewer Service	1	\$94.97	\$94.97
Residential Stormwater Service	1	\$27.00	\$27.00
Residential Water Service	1	\$21.82	\$21.82
Total Service Charges			\$143.79
Usage Charges -Water meter readings over the page			
	Usage (kL) or No. of Charges	Tarrieff or Charge	Amount
Residential Sewer Service	1	\$27.19	\$27.19
Water Usage	72	\$2.10	\$151.20
Total Usage Charges			\$178.39
Total Account Charges			\$322.18

Richard reduced his water usage to 60 kL in the next three-month period (July to September), and all other charges remained unchanged.

What will his Total Account Charges be for this period?

- (A) \$151.99
- (B) \$205.66
- (C) \$296.98
- (D) \$320.18

8. A medical experiment is conducted on a sample of 1 500 members of the public to identify if they showed the symptoms of a certain syndrome.

The results were classified into five increasing levels of symptoms and the relative frequencies for these are displayed in the table below.

Outcome	Relative Frequency
No Symptoms	0.64
Indeterminate Symptoms	0.21
Mild Symptoms	0.06
Significant Symptoms	0.02
Severe Symptoms	0.07

If the experiment is valid for all members of the population, how many people in a city of 85 000 people would show mild through to severe symptoms of the syndrome?

- (A) 12 750
- (B) 17 850
- (C) 30 600
- (D) 54 400
9. Fried's formula, below, is used to calculate the infant dosages of medicines.

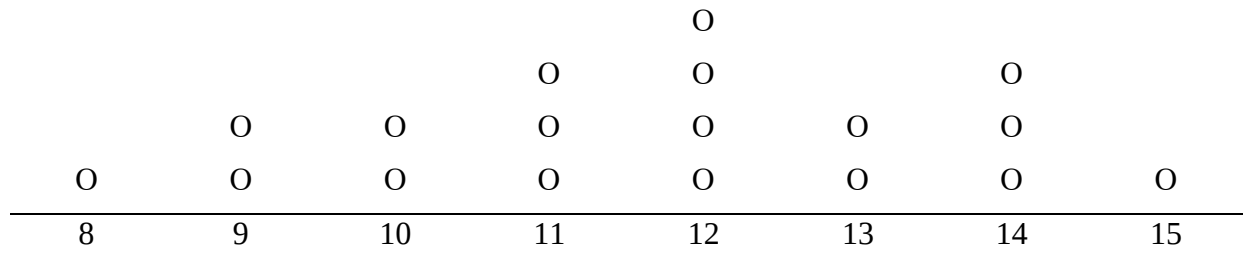
$$\text{Dosage for children 1 – 2 years} = \frac{\text{age (in months)} \times \text{adult dosage}}{150}$$

Naomi uses Fried's formula to calculate the dosage of an antibiotic for a child who is 1 ½ years old to be 6 millilitres.

What was the adult dosage for the antibiotic?

- (A) 30 millilitres
- (B) 50 millilitres
- (C) 60 millilitres
- (D) 120 millilitres

10. The dot plot below shows the number of tubes of paint in the kits of a sample of 18 artists.



What is the interquartile range of the dataset?

- (A) 3
- (B) 3.5
- (C) 4
- (D) 7

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Western Mathematics

**2022 YEAR 11 FINAL
EXAMINATION**

Class and Teacher

Mathematics Standard

Student Number

Section II Answer Booklet

Student Name

70 marks

Attempt Questions 11 – 36

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

Solve the equation: $10 - \frac{k}{3} = k + 2$.

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Question 12 (2 marks)

Patrick measures the mass of a bag of potatoes as being 5.0 kilograms, using a scale which has markings every 0.2 kg.

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Calculate the upper and lower bounds for the limits of accuracy of this measurement.

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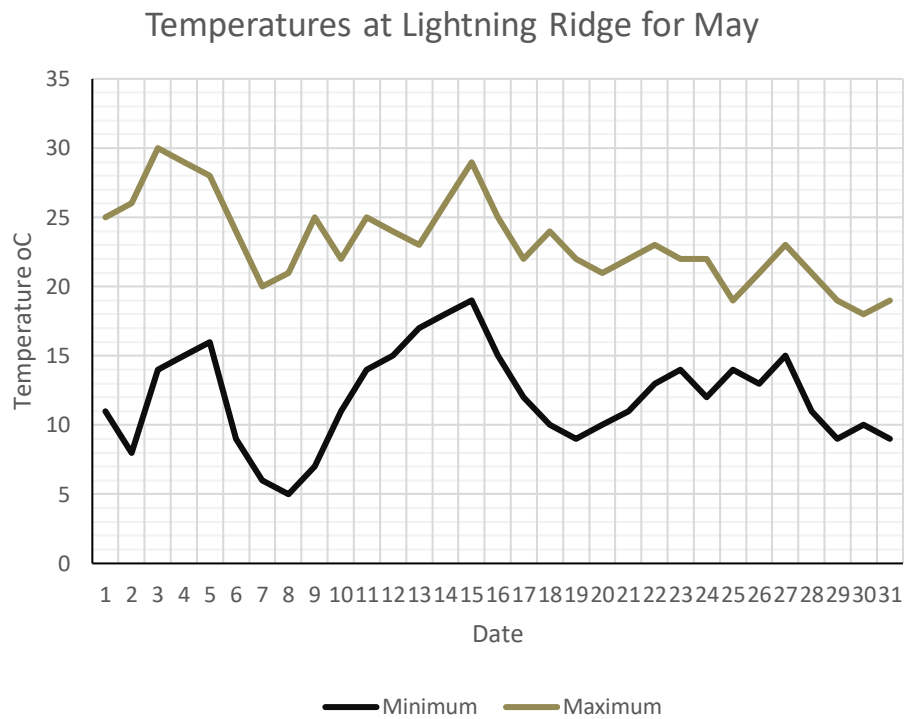
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Question 13 (2 marks)

The graph below gives the daily maximum and minimum temperatures for May in Lightning Ridge NSW.



- (a) What is the difference between the highest maximum and lowest minimum temperatures for the month?

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- (b) Describe any trend in the maximum temperature recordings over the month.

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Question 14 (2 marks)

Given the formula $s = ut + \frac{at^2}{2}$ find the value of u when $s = 195$, $t = 5$, and $a = 10$.

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Question 15 (2 marks)

Cassie works at a radio station where her normal rate of pay is \$46.40 per hour for the first 36-hours each week.

2

She is paid overtime at time and a half for the next 4 hours per week and double time after that.

What would she be paid for the week shown in the timesheet below?

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Hours	7	8	7	8	7	0	4

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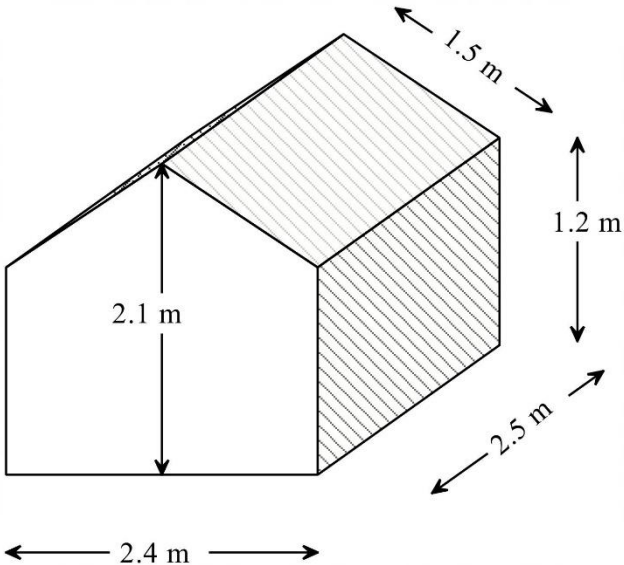
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Question 16 (2 marks)

A tent has the dimensions shown below and is made of lightweight fabric.
The tent has no floor included.

2



Find the area of fabric required for the tent.

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Question 17 (2 marks)

In a game of chance, a person invests \$2 and can win \$4, \$8 or \$16 or they can lose their stake.
The probability of each is shown below.

2

Outcome	Lose Stake	Win \$4	Win \$8	Win \$16
Probability	$\frac{16}{25}$	$\frac{7}{25}$	$\frac{3}{50}$	$\frac{1}{50}$

What is the probability that a person will not win either of the two highest amounts?

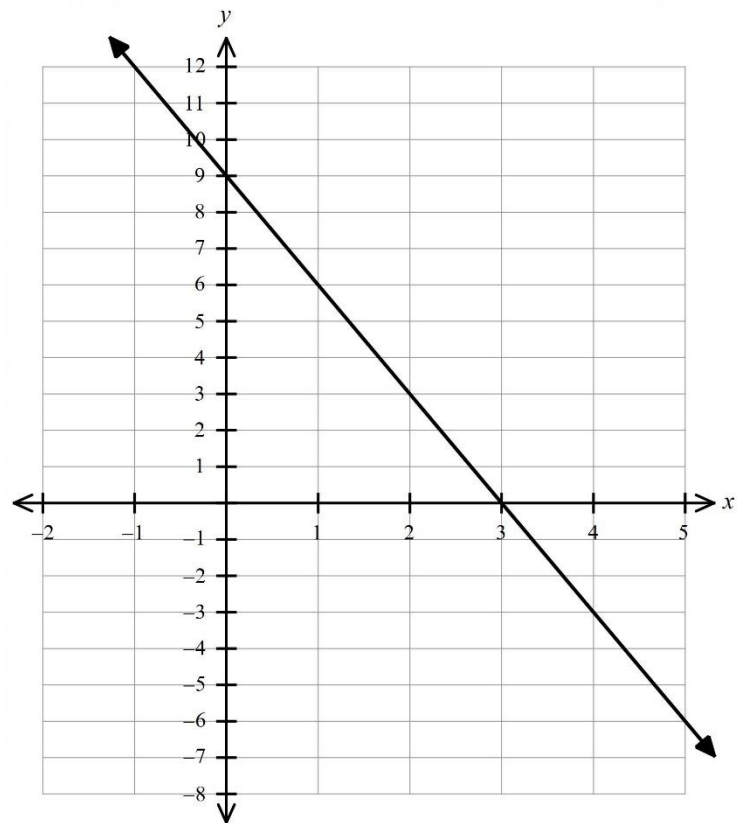
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Question 18 (3 marks)

A linear graph has been drawn on the axes below.



(a) What is the gradient of the line shown?

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(b) On the same axes, draw the graph of $y = 4x - 5$.

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(c) Give the coordinates of the point of intersection of the two graphs.

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Question 19 (4 marks)

A frequency distribution table is shown for a dataset.

At class boundaries, values on the boundary fall into the lower class.

Class	Class Centre	Frequency	Cumulative Frequency
2.5 – 7.5	5	12	
7.5 – 12.5	10	18	
12.5 – 17.5	15	40	
17.5 – 22.5	20	45	
22.5 – 27.5	25	55	
27.5 – 32.5	30	30	

(a) Complete the Cumulative Frequency column. 1

(b) What is the modal class of the dataset? 1

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(c) Find an estimate for the mean of the dataset. 1

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(d) Describe the shape of this distribution. 1

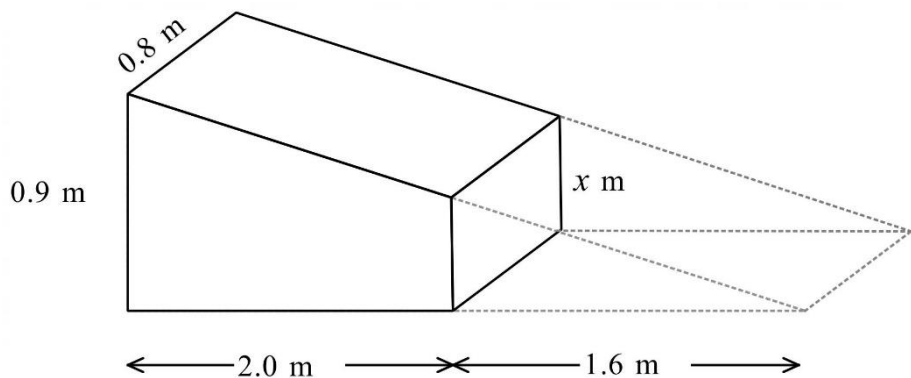
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Question 20 (3 marks)

The diagram below shows a steel trapezoidal prism which is formed by cutting a section off a triangular prism.



- (a) Use similar triangles to show that $x = 0.4$ and hence find the volume of the trapezoidal prism in cubic metres.
- 2

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- (b) Given that the density of steel is $7\,850\text{ kg/m}^3$, find the mass of the prism in tonnes, to the nearest 10^{th} of a tonne.
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Question 21 (2 marks)

Zoe buys an electric space heater which is rated at 1 500 watts. She is on a plan where the price for electricity is \$0.234 per kWh.

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What will it cost her to run the heater continually for 24 hours?

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Question 22 (3 marks)

Brydon created the spreadsheet below to keep track of his spending on Transport over a year.

	A	B	C	D	E	F	G
1							
2		Transport Budget					
3			Amount	Frequency	Payments	Total	
4		Vehicle repayments	\$225.40	Monthly	12	\$2,704.80	
5		Bus fare	\$35.50	Weekly	52	\$1,846.00	
6		Insurance	Y	Monthly	12	\$385.20	
7		Registration	\$856.00	Annual	1	\$856.00	
8		Fuel	\$86.00	Weekly	52	\$4,472.00	
9		Maintenance	\$325.00	Quarterly	4	\$1,300.00	
10							
11		Total				Z	
12							

Two cells have been blanked out in the spreadsheet and labelled Y and Z.

(a) What are the values in the cells labelled Y and Z?

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Question 22 continues on page 18

Question 22 (continued)

- (b) Brydon wants to put a weekly amount into an account that will, over the year, cover all his transport costs.
How much should he deposit in the account each week?
Answer to the nearest 10 cents.

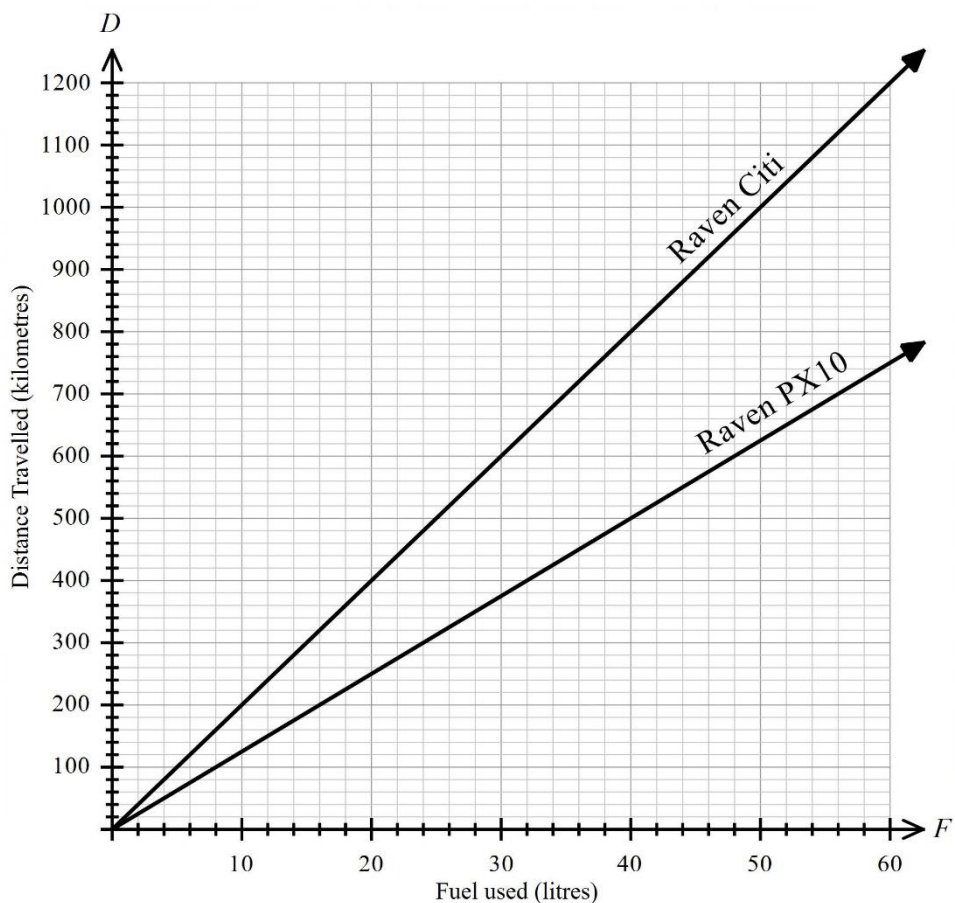
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Question 23 (4 marks)

There is a direct variation relationship between the amount of fuel used by a vehicle and the distance which could be travelled using that fuel.

The graph below shows this relationship for two different models of car produced by the Raven motor company.



Question 23 continues on page 19

Question 23 (continued)

- (a) How many extra kilometres could be travelled in the Citi model compared to the PX10, if there were exactly 40 litres of fuel in the tank of each model? **1**

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- (b) A third model, the Raven Tourer, uses 10L per 100 km. **2**
Draw and label the line for this model on the graph on page 18.

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- (c) All three models are sent on a 600 km test drive. **1**
What is the difference between the highest and lowest amounts of fuel used by the three models?

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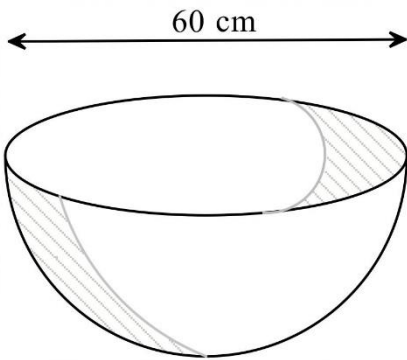
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Question 24 (2 marks)

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A bowl is a hemisphere as shown below.



Given that 1000 cm^3 holds one litre of water, how many litres of water would the bowl hold? Answer to the nearest litre.

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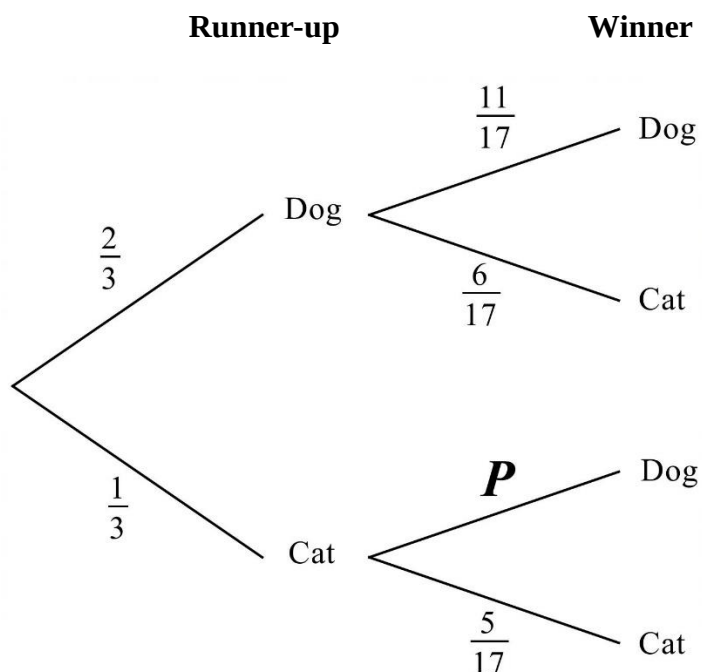
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Question 25 (3 marks)

In a Prettiest Pet competition at a school fete there are 18 pets, one third of which are cats, with the rest being dogs. There are two prizes, and the Principal must choose the runner-up, followed by the winner. He cannot decide, so he chooses two pets at random.

The tree diagram below has been started to show the possibilities.



- (a) What is the missing probability ***P***? **1**

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- (b) What is the probability that a dog is runner-up, and a cat is the winner? **1**

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- (c) What is the probability that the runner-up and the winner are the same species? **1**

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Question 26 (2 marks)

Anne is paid an annual salary of \$53 049 after tax, and her partner Richard has an annual salary of \$44 980 after tax, and both are paid weekly. They have calculated their weekly combined expenses (rent, transport, food, entertainment etc) to be \$1 643.00. They decide to save all the money which is not used for these expenses.

2

How much could they save each week, and what would this add up to over a year?

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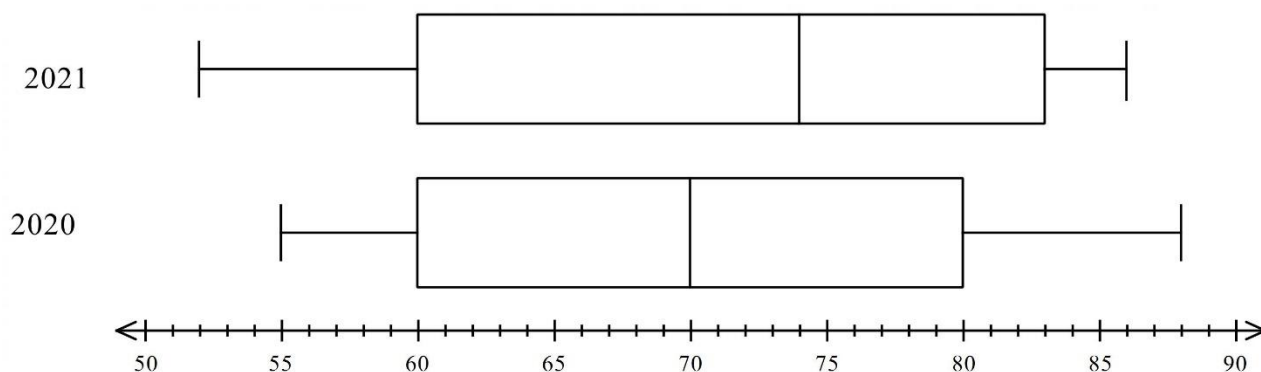
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Question 27 (3 marks)

The parallel box-plots below shows the comparison between the HSC marks on Standard Mathematics over two years at Danville High.



HSC Marks in Standard Mathematics at
Danville High

Question 27 continues page 23

Question 27 (continued)

- (a) Compare the spread of the data for the two years using two different statistical measures. 2

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- (b) A teacher claimed that the overall results in Standard Mathematics had improved in 2021, compared to the previous year. 1

Describe how this claim is supported by statistical measure(s).

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Question 28 (3 marks)

Rearrange the formula $B = P + \frac{PRN}{100}$ so that the subject is R . 3

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Question 29 (2 marks)

Kerri buys a photocopier for \$15 600 for her office. It is depreciated, using the straight-line method, by 1.25% of its original value each month. It can be replaced when its value falls below \$5 000.

2

After how many months could it be replaced?

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Question 30 (4 marks)

Jamie has a gross annual income of \$98 020.00 from his job and interest income of \$560.00.

He has the following allowable deductions.

- Work equipment and clothing expenses - \$2 540.00
- Gifts and donations - \$3 200.00
- Home Office expenses - \$895.00
- Travel expenses - \$3 453.00

(a) What is his taxable income?

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Question 30 continues page 25

Question 30 (continued)

- (b) Use the table below to calculate the tax that Jamie should pay, including the Medicare Levy.2

Resident tax rates 2021–22

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$45,000	19 cents for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5 cents for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37 cents for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45 cents for each \$1 over \$180,000

The above rates **do not** include the Medicare levy of 2%.

- (c) During the tax year he paid \$825.60 per fortnight in PAYG tax instalments.1
Determine the amount of his tax refund or debt.

Question 31 (3 marks)

A game produces random avatars for players of the game.

The array below shows how two of the characteristics of the avatars are distributed.

		Hair colour					
Avatar Type		Red	Green	Yellow	Purple	Ebony	Bald
	Male Dwarf						
	Female Dwarf						
	Male Elf						
	Female Elf						
	Male Hobbit						
	Female Hobbit						

All the characteristics are equally likely.

- (a) What is the probability that a player will receive a purple-haired female elf as their avatar?1

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- (b) What is the probability that the avatar will be green-haired or a hobbit, or both?1

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- (c) What is the probability that the avatar will not be a bald dwarf?1

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Question 32 (2 marks)

Part of a tram timetable is shown below for Saturday mornings.

2

Stop	Sat am	Sat am	Sat am	Sat am	Sat am
Valentine St	6:33	7:03	7:33	7:44	8:06
Glover St	6:38	7:08	7:38	7:49	8:11
Armstrong St	6:45	7:15		7:56	
Mccullough St	6:51	7:21	7:50	8:02	8:23
Mungindi St	6:54	7:24		8:05	
Lane Ln	6:59	7:29	7:57	8:10	8:30
Chapppel Av	7:07	7:37		8:18	
Sara St	7:16	7:46	8:13	8:27	8:46
McNamara St	7:21	7:51		8:32	
Karvalho Rd	7:27	7:57		8:38	
Sales Pl	7:31	8:01	8:25	8:42	8:58
Corowa Sq	7:37	8:07	8:31	8:48	9:04
Windradyne Rd	7:42	8:12	8:36	8:53	9:09

Martin, who lives 15 minutes from the Armstrong St tram stop, needs to get to an appointment at 9:00 am on Saturday, in a building which is 10 minutes-walk from the Carvalho Rd tram stop.

If all trams run on time, what is the latest time that he can leave home to make the appointment on time?

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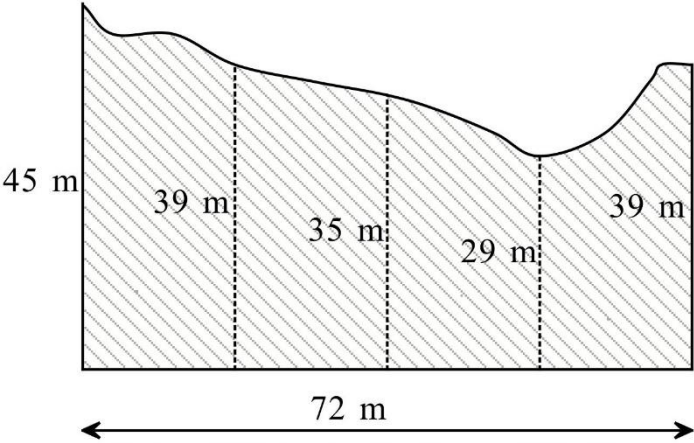
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Question 33 (2 marks)

2

The field shown below has three straight boundaries and one curved boundary which follows a creek.

To estimate the area of the field, offsets are taken at regular intervals from the 72 m boundary and their lengths are measured.



Apply the trapezoidal rule, using four trapezia, to estimate the area of the field.

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Question 34 (4 marks)

A machine produces concrete gutters for urban streets.
The volume of concrete required (v) varies directly with the length of gutter produced (L).
The relationship between v and L can be expressed as $v = kL$, where k is a constant.

- (a) Given that 6 m^3 of concrete produces 40 m of gutters, find the value of k . 1

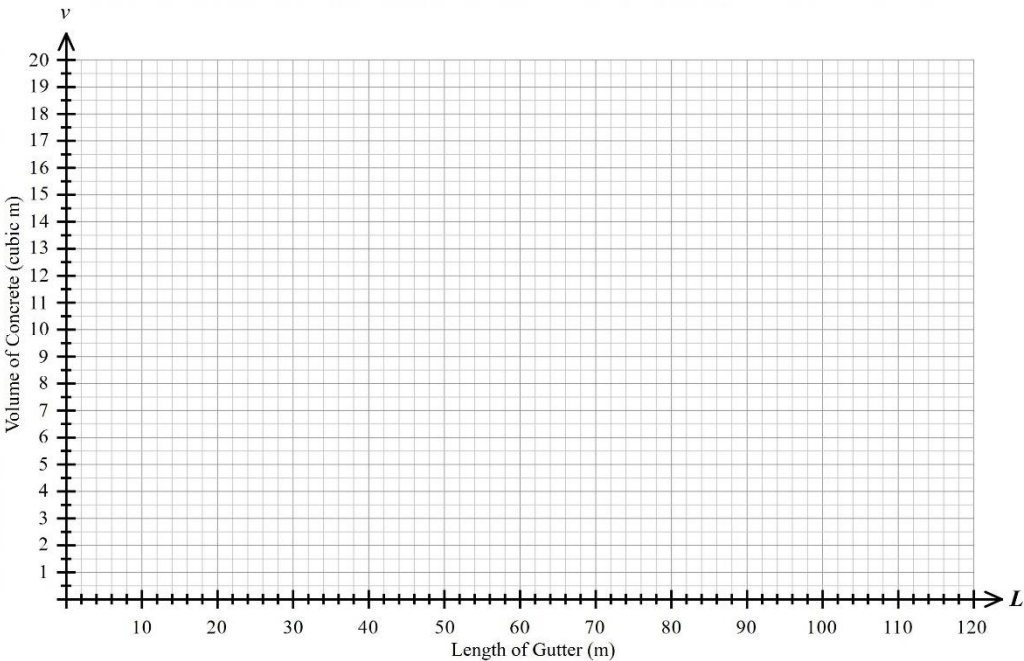
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- (b) Use the axes below to draw a graph illustrating the relationship between L and v . 2



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- (c) The maximum capacity of the machine is 16.5 m^3 of concrete. 1
What is the longest gutter that could be laid before the machine needed refilling?

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Question 35 (3 marks)

Kate is at Sydney airport which is UTC +10 and is about to fly to Honolulu in Hawaii, which is UTC −10.

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It is 4:00 p.m. on Friday, Sydney time, when her flight leaves.

The flying time to Honolulu is 6 hours, and there is a 3 hour stop in Fiji on the way.

Kate calls her mum in Sydney, when she reaches her hotel, 3 hours after landing.

What time does she make the call Honolulu time and what time does her mum receive it, Sydney time?

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Question 36 (4 marks)

Dominica goes over her accounts for last year and works out how much of her income was spent in four broad sectors of expenditure. She was not able to save any of her income last year.

Expenses Last Year	
Sector	Annual Amount
Accommodation	\$31 200
Transport	\$5 980
Food and Groceries	\$11 440
Daily and Discretionary Spending	\$10 140
Total Income	\$ 58 760

She wants to plan a budget for this year.

She knows that her income will increase by 2.5% over last year's amount.

She wants to be able to save \$5 200.00 by the end of this year.

The Accommodation and Transport sectors will have an unavoidable increase of 3.5% compared to last year.

She decides to keep her Food and Grocery spending at the current rate by being more careful on her spending.

Dominica plans to reduce her Daily and Discretionary Spending as required to achieve her saving goal.

Question 36 continues on page 32

Question 36 (continued)

Complete the calculations necessary and complete the table below for her planned budget for this year.

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Budget for This Year		
Sector	Annual Amount	Weekly Amount
Accommodation		
Transport		
Food and Groceries	\$11 440	\$220
Daily and Discretionary Spending		
Saving	\$5 200	\$100
Total Income		

End of Paper

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

2022 YEAR 11 Final Examination
Mathematics Standard

REFERENCE SHEET

Measurement

Limits of accuracy

$$\text{Absolute error} = \frac{1}{2} \times \text{precision}$$

$$\text{Upper bound} = \text{measurement} + \text{absolute error}$$

$$\text{Lower bound} = \text{measurement} - \text{absolute error}$$

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

An outlier is a score

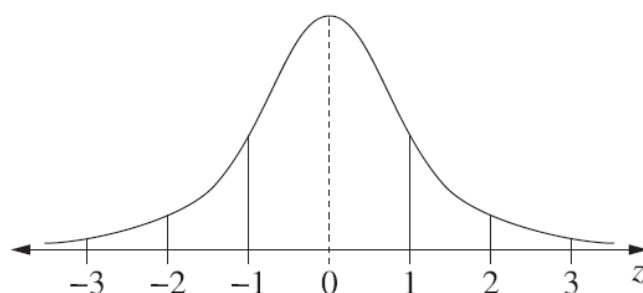
$$\text{less than } Q_1 - 1.5 \times IQR$$

or

$$\text{more than } Q_3 + 1.5 \times IQR$$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution



- approximately 68% of scores have z-scores between -1 and 1
- approximately 95% of scores have z-scores between -2 and 2
- approximately 99.7% of scores have z-scores between -3 and 3

Western Mathematics
2022 Year 11 Final Examination
Mathematics Standard

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |